

The Closure Challenge

A post-hoc RANS correction entry, its verification record,
and the field-inversion line behind it

Certonomous CFD Laboratory

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Status. Nothing has been submitted to the Closure Challenge. No account has been created and the benchmark steward has not been contacted. The entry described here (overall **0.0654**, round 4) is an internally measured position against the public leaderboard, not an official ranking. The submission package exists, is audited, and is one email away; the decision to send it rests with Katie Kim.

Abstract

This report records what the Certonomous CFD laboratory has done in the Closure Challenge, a public benchmark for machine learning in turbulence modelling stewarded by Ryley McConkey (MIT). Our entry of record scores **0.0654** on the benchmark’s own unmodified scorer, which would place it third of five against the current public leaderboard, between Wu & Zhang (0.0624) and Liu, Wang, Zhao & Xiao (0.0737), and 36.9% below the 0.1036 scored by submitting the supplied RANS fields unchanged. The method is a post-hoc velocity field correction — the only entry on the board that does not re-solve the governing equations with its correction folded in — combined with a train-only *decline gate* that withholds the correction on whole cases where it predicts the correction would do harm, and a pre-registration discipline under which every model choice is frozen in a committed document before the prediction set it selects is ever scored. We state the costs of that method class as plainly as its benefits: our submitted fields do not satisfy continuity, and as of yesterday the departure is measured rather than unmentioned — about 10% of the local velocity-gradient scale on the two corrected hills and 2.3–3.4% on the ducts, against baselines that are divergence-free to the operator floor; two of the five leaderboard-best rows we hold are the organisers’ own baseline rather than our model; and one of the remaining three is a lead of 0.00003 that must not be described as a comfortable win. The report also documents the parallel field-inversion line, which in the last week moved from a hard numerical blocker (`PetscConvergedReason: -9`, reproduced on two cases and five matrix orderings) to the laboratory’s first converged, finite-difference-verified closure-relevant adjoint gradient: 21,000 design variables on the curved backward-facing step, agreeing with central differences to 0.021–0.199% on four independently chosen cells.

Provenance convention. Every quantitative claim below is followed by the repository artifact it comes from. Repository paths are relative to the `Certonomous` repository root and are cited with the commit at which the cited text stands; external work is cited by DOI or arXiv identifier and was read from a copy held under `docs/papers/`. Appendix A is the full provenance index. Where a number is a bound rather than a measurement, or an inference of ours rather than a claim of the source’s, this report says so at the point of use.

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1 The challenge, and the metric it is scored on

1.1 What the benchmark is

The Closure Challenge is an open, ongoing benchmark task for machine learning in turbulence modelling. Its authoritative surface is the repository <https://github.com/rmcconke/closure-challenge-benchmark>, held locally at commit `deb91557`; the evaluation package `closure-challenge` is pinned at commit `1c4e22c8`; the accompanying preprint is arXiv:2603.28884 (McConkey, Buchanan, Smidt, Bodner, Dwight, Cinnella), DOI 10.48550/arXiv.2603.28884. The README states in its own words that it, not the paper, is the current source of information, and our records follow that ordering: no rule in this document is sourced to the preprint where the README states the same thing (`CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md` §1–§2, commit `15530f97`).

The task is to predict the velocity field on eight held-out test cases given a suggested training and validation split and a supplied CFD mesh. The eight scored cases are four parametric periodic hills (`alpha_15_13929_4048`, `alpha_15_13929_2024`, `alpha_05_4071_4048`, `alpha_05_4071_2024`), three rectangular ducts (`AR_1_Ret_360`, `AR_3_Ret_360`, `AR_14_Ret_180`) and the NASA wall-mounted hump (`NASA_2DWMH`).

Three properties of the rules matter to everything that follows.

1. **There is exactly one strict rule**, and it is about data, not method: it is “*strictly forbidden to train or validate on any data from the test cases*”, on pain of automatic withdrawal and a permanent note on the leaderboard. The preprint adds the permission the entry leans on, verified verbatim from the preprint’s own §2.1 on 2026-08-02: “*You can train on similar flows to the test cases.*”
2. **Everything else is unrestricted**. Input features, base turbulence model and data-assimilation technique are explicitly the submitter’s choice; the published train/validation split is labelled “(*suggested*)” and only the test column binds.
3. **There is no scoring-call limit, and there cannot be one**. The evaluation package ships the test ground truth locally and the README instructs submitters to preview their score; `evaluate_by_case()` is documented public API. Our five-calls-ever ledger (§6.2) is therefore a self-imposed discipline, *stricter* than the benchmark requires, and must never be described as compliance with a rule that does not exist.

There is no eligibility clause of any kind in the benchmark README, the evaluation package README or the preprint — an absence of restriction, reported here as an absence rather than as a permission. No fee, registration, account or affiliation is required; submission is an email of a `test/` directory to the steward.

1.2 The metric, read from the evaluation code

The metric was read directly from `closure_challenge/eval.py` rather than from our own prose (`CLOSURE_METHODS_COMPARISON.md` §1.0, commit `ac2f37ee`). For a case with 1000 fixed evaluation points, `evaluate_individual_case()` computes a scaled mean absolute error of the *vector magnitude* of the velocity error,

$$s_{\text{case}} = \frac{\frac{1}{N} \sum_{i=1}^N \|\mathbf{U}_i^{\text{pred}} - \mathbf{U}_i^{\text{true}}\|_2}{\frac{1}{N} \sum_{i=1}^N \|\mathbf{U}_i^{\text{true}}\|_2}, \quad N = 1000, \quad (1)$$

and `score()` returns the plain, unweighted mean of the eight per-case values. Lower is better. Two consequences are worth carrying: the error is a vector-magnitude error and not component-wise, and because the mean is unweighted, an improvement of Δ on one case moves the overall

by $\Delta/8$ — so the two `alpha_15` cases, whose baseline errors of 0.1320 and 0.2049 dominate the floor, carry exactly the same $1/8$ weight as everything else.

All scoring in this project is done through the benchmark’s own unmodified scorer. The metric is never reimplemented for scoring purposes.

1.3 The public leaderboard, and where we stand

Table 1 gives the public board as published at benchmark commit `deb91557`, with our unsubmitted entry inserted at its measured position. All four published entrants re-score to their published values exactly on our harness, per-case and overall, which is the strongest available evidence that the harness is being driven correctly (`CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md` §7.5).

Table 1: The public leaderboard, with our entry of record at its internally measured position. We are unsubmitted; this is not an official ranking.

Rank	Entry	Overall
1	Reissmann, Fang & Sandberg	0.0595
2	Wu & Zhang	0.0624
—	Certonomous (unsubmitted, round 4, entry of record)	0.0654
—	Certonomous (unsubmitted, round 3, superseded)	0.0676
3	Liu, Wang, Zhao & Xiao	0.0737
4	Montoya, Oulghelou & Cinnella	0.0779
—	RANS-identity floor (submit the supplied fields unchanged)	0.1036

Two cautions attach to that table and are repeated wherever it appears.

- **The 0.1036 floor is our own number.** The script that produces it (`scripts/rans_identity_baseline.py` in the benchmark clone) is untracked — it was written by this laboratory, not shipped by the benchmark, and the README quotes no RANS-identity baseline anywhere. It reproduces exactly and is honestly derived, but any wording implying the floor is the benchmark’s own published figure is false (`CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md` §7.2).
- **Position claims go stale.** The challenge is ongoing with no deadline; the board has been unchanged since 2026-05-04 but can change without notice. Every position claim in this document is dated 5 August 2026.

2 Our approach

2.1 Method class, stated at its real strength

Our correction is a **post-hoc velocity-field correction**. The learning target is $\delta\mathbf{U} = \mathbf{U}_{\text{LES}} - \mathbf{U}_{\text{RANS}}$ per mesh cell, and the learned map is applied once to an already-converged RANS field as a final post-processing step. **Nothing re-enters the governing equations and nothing is re-solved with the correction folded in.** In the Duraisamy–Iaccarino–Xiao taxonomy this is the “correct the answer” family, structurally distinct from FIML, TBNN, SpARTA and eigenvalue perturbation, all of which correct something inside the equations before re-solving (`docs/research/CLOSURE_METHODS.md`; `CLOSURE_CHALLENGE_STATUS.md` §4, commit `b9cb56a2`).

Two model objects stand behind the six corrected predictions of the entry of record (`CLOSURE_METHODS_COMPARISON.md` §1.1):

- **The periodic-hill (PH) model** — three `HistGradientBoostingRegressors`, one per velocity component, `max_iter=300`, `max_depth=6`, `learning_rate=0.05`, `l2_regularization=1.0`,

`random_state=0`, on seven per-cell features: Pope’s five scalar invariants of the normalised strain and rotation tensors (`I1_S2`, `I2_W2`, `I3_S3`, `I4_W2S`, `I5_W2S2`; Pope, *J. Fluid Mech.* 72 (1975) 331–340), a wall-distance turbulent Reynolds number `Re_y`, and a turbulent-to-mean kinetic energy ratio `tke_ratio`. Trained on 21 periodic-hill cases, 327,600 cells.

- **The duct model, “Variant D”** — the same architecture on eight features (adding a Reynolds-invariant $d/d_{\max}$) with a velocity-scale-normalised target, $(\mathbf{U}_{\text{LES}} - \mathbf{U}_{\text{RANS}})/|\mathbf{U}_{\text{RANS}}|$ per case, trained on the four suggested training ducts, 41,971 cells.

Round 2 required building a feature pipeline the benchmark does not ship for the duct and CBFS families: Green–Gauss $\nabla \mathbf{U}$ reconstructed from mesh, RANS $\mathbf{U}$ and boundary conditions, and wall distance from the nearest wall-patch face centre. It was validated *only* against shipped fields on periodic-hill *training* cases: cell centres and volumes to $\sim 1\text{e-}13$ and $\sim 1\text{e-}22$ absolute, $\nabla \mathbf{U}$ to a pooled relative Frobenius error of 0.0012–0.0029 with Pearson $r = 0.9997$ –1.0000 on three different training cases, and wall distance to $r = 1.0000$ with mean relative error below 0.1% (`CLOSURE_CHALLENGE_STATUS.md` §4).

2.2 The do-no-harm gate

The distinguishing component of the entry is a **case-level decline gate**. It is a three-feature ridge regression (top-3 screened features `p90_I4_W2S`, `frac_backflow`, `p90_I3_S3`; RidgeCV $\alpha = 0.7499$) that predicts, from RANS-derived quantities alone, the scaled MAE the *uncorrected baseline* will achieve on a case. If the predicted baseline error falls below a threshold of 0.1263 — the leave-one-out median over the training cases — the correction is withheld and the unmodified RANS field *is* the submission for that case (`closure_challenge_C1_error_estimator.md`, commit 5719374e).

Every parameter of the gate — feature screening, ridge α , standardisation, final coefficients and threshold — was computed from the 21 periodic-hill training cases and nothing else, and was validated on the four suggested validation cases (AUC 1.0, 4 of 4 correct) before it was ever pointed at a test case. When it was applied, the refit reproduced $\alpha = 0.7499$ and threshold 0.1263 byte-for-byte from train-only data as a precondition of the run.

Two honest qualifications belong beside it:

- **Its statistical base is thin, and this was disclosed before it was useful.** AUC 1.0 on $n = 4$ with a single positive is approximately a one-in-four outcome by luck; the gate has made eight binary decisions in total. Widening the validation base to a 25-case leave-one-out is a filed proposal (G5, §6.3).
- **It is not novel, and the nearest prior art is by the challenge’s own authors.** Classifier-controlled data-driven corrections are established: Ling & Templeton, *Phys. Fluids* 27, 085103 (2015); Wu, Wang, Xiao & Ling, *Flow Turbul. Combust.* 99, 25 (2017), DOI 10.1007/s10494-017-9807-0; Steiner, Dwight & Viré, *Flow Turbul. Combust.* (2022), DOI 10.1007/s10494-022-00346-6; and Buchanan, Lăcătuș, West & Dwight, *Computers and Fluids* 305, 106899 (2025), DOI 10.1016/j.compfluid.2025.106899, arXiv:2504.06758 — whose first and last authors are co-authors of the Closure Challenge paper itself. Read in full, their RITA classifier selects *regions inside a case* from local ratios of the baseline’s own k -equation terms. What survives of our distinction, and all we claim, is: our gate acts on a *whole case*; it predicts the *baseline’s error* rather than distribution shift; and its “off” state means the uncorrected field *is* the submission (`CLOSURE_CHALLENGE_PRIOR_ART.md` §2.4, §7.4).

That last reading carried a constraint we wrote down before anyone felt the pull of it: Buchanan et al. train on the NASA wall-mounted hump, which is one of our eight scored test cases. Their published model coefficients are therefore **off limits** to this entry — borrowing them

or calibrating anything of ours against that model’s hump behaviour would make our submission indirectly trained on a test case.

2.3 Pre-registration

Every model choice that selects between candidate prediction sets is frozen in a committed document *before* the prediction set it selects is scored. The NASA hump’s model choice was pre-registered before the round-2 scoring call; Variant D was pre-registered on the benchmark’s own suggested duct validation case `AR_7_Ret_180` before any test score for it existed (`closure_challenge_duct_reynolds_transfer.json`, commit `d0992da2`); the round-5 candidate wrote down its accept/reject rule, then *failed its own pre-declared gate and was not scored* (§6.5).

Pre-registration that only ever ratifies is decoration. The R5 no-go is the evidence that this one binds.

2.4 The costs of this method class, stated before the results

Table 2 compares the five entries dimension by dimension, from each entrant’s own documents (`CLOSURE_METHODS_COMPARISON.md` §3.1). Read the top two rows first, because they are the ones that cost us.

Table 2: The five entries, dimension by dimension. “Absent from their documents” means absent from the sources read — two submission documents, one notebook plus info file, and two full papers — and is stated as an absence in those sources, not as a claim about any laboratory’s unpublished practice.

Dimension	Certonomous	Reissmann	Wu & Zhang	Liu et al.	Montoya et al.
Corrected object	velocity field, post-hoc	anisotropy + SST coefficients	β on ω -destruction + fixed QCR	TBNN anisotropy + k - ω coefficients, coupled	blended tensor-basis + R_k
Re-solved with the correction	no — unique among the five	yes (inferred)	yes	yes	yes
Conservation satisfied by the submitted field	no — and now measured (§6.4)	yes, by construction	yes	yes	yes
Trains on a scored case	no — enforced by executable asserts	no	no	yes, per their own paper (entry-level status unverifiable)	no
Case-level do-no-harm gate	yes	absent from their documents	region shield f_d , hand-designed	absent from their documents	baseline-as-expert local fallback
Pre-registration before scoring	yes	absent	absent	absent	absent
Scoring-call ledger	yes, 5 calls, self-imposed	none visible	none visible	none visible	none visible

The first three rows are the honest headline of this report. **Every entrant above us ships a method whose output satisfies the governing equations by construction; ours does not.** The submitted corrected field is a cell-wise machine-learning output added to a solenoidal RANS solution, and it does not satisfy continuity by construction. No realizability question

arises — we predict velocity, not Reynolds stresses, so that column is not-applicable rather than passed — but $\nabla \cdot \mathbf{U}$ is a real property of a submitted field and until this week nobody had measured it. It has now been measured against a threshold fixed before the run, and the finding is material and against us (§6.4).

The compensating claim is narrow and we keep it narrow: among the five entries on this board, ours is the only one that will decline to correct a whole case and submit the baseline instead, and the only one that freezes its choices in advance of scoring.

3 Results, round by round

3.1 The arc

Round 1 — 0.0869. The PH model, trained on 21 periodic-hill cases and validated on 4, applied to the four periodic-hill test cases only. All four non-PH test cases were passed through as unmodified raw RANS, identical to the floor, because the benchmark does not ship $\nabla \mathbf{U}$ and wall distance in solved time directories for the duct and CBFS families and reconstructing them was out of scope for that round.

Round 2 — 0.0741 ($\Delta -0.0128$). The mesh-reconstruction pipeline of §2.1 unlocked the remaining four cases. A new duct-specific model was trained on `AR_1/3/5/10_Ret_180` and validated on the held-out `AR_7_Ret_180`; `NASA_2DWMH` was corrected with the existing PH model, a pre-registered choice made before test scoring. The hump regressed slightly, from 0.0621 to 0.0632, and was reported rather than reverted.

Round 3 — 0.0676 ($\Delta -0.0065$). The C1 decline gate, built and validated in a prior session on non-test data only, was evaluated on the four periodic-hill test cases’ RANS-derived features for the first time. It declined both `alpha_05` cases (predicted baseline 0.0420 and 0.0034, both below the 0.1263 threshold) and kept the correction on both `alpha_15` cases (0.1503 and 0.1525, both above) — 4 of 4 correct on cases it had never been evaluated on, matching its validation performance exactly. The movement, -0.0065 , matched the -0.0066 predicted by the round-2 error decomposition almost exactly. Compute: 83 s wall on a two-core cap, peak RSS 356 MB.

Round 4 — 0.0654 ($\Delta -0.0022$), the entry of record. A duct-only change. Variant D replaced the round-2/3 duct model on all three ducts. The five non-duct predictions are *byte-identical* to round 3 — copied, not regenerated, and SHA-256-verified against the entry manifest — so the entire delta is attributable to the duct family and the gate’s behaviour is untouched.

Round 4’s diagnosis was pre-registered and its sign pattern was predicted before it was measured. The published duct diagnosis had been *wrong*: `I3_S3` and `I4_W2S` are indeed algebraically zero on every duct, but they are equally zero on `AR_14_Ret_180`, the duct we score best on, so they discriminate nothing. What discriminates is that `Re_y` reaches $1.85\times$ and $2.07\times$ its trained maximum on the two ducts we were losing, against $0.90\times$ on the one we win — and a gradient-boosted tree cannot extrapolate. The pre-registration predicted improvement concentrated on the two extrapolating cases and little or none elsewhere. Measured: -0.0108 , -0.0087 , and $+0.0022$. The sign pattern matches on all three (`CLOSURE_CHALLENGE_STATUS.md` §0e).

3.2 The per-case ledger

3.3 How the “best on board” count must be reported

Arithmetically we hold the best score on the public board on five of eight cases. That count was on our public page as a headline and **we took it down ourselves**, because it is true arithmetic

Table 3: Per-case scaled MAE across rounds, against the RANS-identity floor, rank 2, and the best published score on each case. Lower is better. Round-4 values are the entry of record.

Case	Floor	R2	R3	R4	Rank 2	Best	Attribution
alpha_15_13929_4048	0.1320	0.0501	0.0501	0.0501	0.0813	0.0592	our model leads
alpha_15_13929_2024	0.2049	0.1011	0.1011	0.1011	0.1195	0.1195	our model leads
alpha_05_4071_4048	0.0461	0.0723	0.0461	0.0461	0.0569	0.0569	<i>baseline, declined</i>
alpha_05_4071_2024	0.0719	0.0974	0.0719	0.0719	0.0848	0.0760	<i>baseline, declined</i>
AR_1_Ret_360	0.1288	0.0919	0.0919	0.0811	0.0455	0.0387	behind, 3rd of 5
AR_3_Ret_360	0.1243	0.0862	0.0862	0.0775	0.0399	0.0341	behind, 3rd of 5
AR_14_Ret_180	0.0590	0.0303	0.0303	0.0325	0.0350	0.0325	lead of 0.00003 only
NASA_2DWMH	0.0621	0.0632	0.0632	0.0632	0.0364	0.0364	behind, last on board
Overall	0.1036	0.0741	0.0676	0.0654	0.0624	—	rank 3 of 5

and misleading attribution (CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md §8.2). On alpha_05_4071_4048 and alpha_05_4071_2024 our submitted field *is* the organisers’ own unmodified RANS solve — re-derived independently from the supplied k - ω SST field and mesh and diffed against the shipped CSV to a maximum deviation of 5.0×10^{-10} , which is the CSV’s own write precision. Those two values are properties of a file every entrant is handed for free; anyone submitting it unchanged scores exactly the same. The honest attribution is Table 4.

Table 4: Attribution of the eight test cases, replacing the retired “5 of 8” headline.

Attribution	Count	Cases
Our model, clear lead	2	alpha_15_13929_4048, alpha_15_13929_2024
Our model, nominal lead only	1	AR_14_Ret_180
The baseline led; the gate withheld our model	2	alpha_05_4071_4048, alpha_05_4071_2024
Behind	3	AR_1_Ret_360, AR_3_Ret_360, NASA_2DWMH

What *is* defensible is a better finding than the one it replaces, and it is about the selection rather than the score: crossing the recorded per-case floor against the published per-case leaderboard shows that **on exactly two of the eight test cases the uncorrected baseline beats all four published entries — and they are exactly the two cases the gate declined**. A rule fitted on 21 training cases and validated on 4 non-test cases, having never seen a test case, identified blind the two flows on which every published method in this field damages the answer. That is a finding about the benchmark which the steward may value more than our own number, and the draft cover email says so.

3.4 The AR_14_Ret_180 regression, deliberately not reverted

Round 4 improved two ducts and made a third worse: AR_14_Ret_180 went from 0.0303 to 0.0325, +0.0022. It was not reverted, and the reason is the same reason the NASA hump’s +0.0011 was not reverted in round 2: choosing per case between two models *after* seeing their per-case test scores is selection on test outcomes, which is exactly what the benchmark’s one strict rule exists to prevent. Variant D was frozen on validation evidence and applied to all three ducts or none. It was applied to all three.

Margin warning, stated so it cannot be misquoted. AR_14_Ret_180’s best-on-board lead over Reissmann collapsed from 0.0022 to **0.00003** (ours 0.0324698, against the published four-decimal 0.0325). **That is a nominal lead, not a meaningful one, and it must not be reported as a comfortable win.** The same case carries the entry’s weakest numerical-provenance column, which makes any unmeasured numerical slop on it material.

3.5 Three refusals

Three separate score improvements were identified, priced, and declined — each because taking it would have required choosing what to submit case by case from the answer key.

1. **0.0066, refused in round 2.** Switching the correction off on the three cases where round 2 was worse than doing nothing would have scored 0.0675. Knowing *which* three required reading round-2’s per-case test scores. The honest route — build a train-only gate first, validate it on non-test data, then apply it — recovered 0.0065 of the same term in round 3 and cost 0.0001. The laboratory’s own compliance audit calls this “the most defensible decision in the record”.
2. **The AR_14_Ret_180 revert, refused in round 4** — worth 0.0022 on one case, 0.000275 on the mean.
3. **0.00014, refused in 2026-08-01.** On NASA_2DWMH our correction is worse than not correcting (0.0632 against a floor of 0.0621); submitting the untouched solve there is one file copy, breaks no benchmark rule, and would move the overall from 0.06543 to 0.06529. The only reason to single out that one case is that the benchmark told us its score. It is the smallest of the three refusals and it was refused on identical grounds (CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md §8.3).

4 Where the deficit lives, and what separates us from rank 2

4.1 The decomposition, and the mechanism it found

The C2 record decomposes the round-2 result case by case (`closure_challenge_C2_error_decomposition.md`, commit 1da013a8). Its sanity check ran first: recomputing the eight-case means from the per-case values reproduces the published floor 0.1036 and entry 0.0741 exactly.

Its headline was uncomfortable and is the reason the gate exists: **on three of eight cases our correction was worse than doing nothing**, and the two `alpha_05` regressions (+0.0262 and +0.0255) were an order of magnitude larger than the already known NASA hump regression (+0.0011), accounting for 98% of the 0.0528 of summed damage.

The first explanation offered — an “alpha-regime failure” — was replaced by the same document. Two facts kill it: the training split already contains five `alpha_05` cases, with unremarkable per-case training MAE of 0.0589–0.0719, so the model was not extrapolating into an unseen regime; and NASA_2DWMH is not a periodic hill at all yet sits in the same degraded group. The framing that survives is **the correction hurts where RANS was already good**: ordering the five PH-model cases by baseline quality separates them without overlap, hurting at every floor ≤ 0.0719 and helping at every floor ≥ 0.1320 , a gap of 0.0601 with nothing in it, Pearson $r = -0.9418$. *Caveat kept from the source: $n = 5$. A correlation of -0.94 on five points is suggestive, not established.* What earns it precedence over the alpha framing is not the r value but that it explains all three regressions with one mechanism, including the one the alpha story cannot touch.

4.2 Self-correction as a working practice

C2 is the most-corrected document in the closure line, and this report presents that as a feature rather than an embarrassment. Table 5 lists its corrections in place, each dated, each left visible in the record beside the text it corrects rather than silently rewritten.

The same practice runs through the rest of the record: a partial withdrawal of the sentence “no alternative training-family model exists for NASA_2DWMH” once CBFS was found shipped as legal training data (`closure_challenge_C6_hump_decision.md`, commit 1f5e2403); the retirement of

Table 5: The C2 record’s corrections, all made in place and left visible. The docket describes C2 as “thrice-corrected”.

Date	What was corrected	What replaced it
in document	The “alpha-regime failure” framing	Baseline quality: the correction hurts where RANS was already good. Retained with its $n = 5$ caveat.
addendum	The floor was the wrong yardstick. Measuring against “do nothing” made the ducts look like wins (−28.6%, −30.7%)	Measured against the published board, AR_1 and AR_3 together are 62.9% of the gap to rank 2, and the two alpha_05 cases are the <i>smallest</i> recoverable terms (18.9% combined). The earlier recommendation to target alpha_05 is superseded by the document that made it.
2026-08-01 (ruling R6)	“Reproduce Wu, Zhang & Zhang’s β on the ω -destruction term”	DAFoam exposes betaFI0mega_ on the ω equation’s production term (DAk0megaSST.C:743); the destruction term’s β is the F1-blended model constant with no field hook. The two are not equivalent, because the other ω terms do not scale with β . What this stack can run today is a production-term inversion, and it is not a like-for-like reproduction.
2026-08-04 (W2 deep read)	One paper cited where there are two artifacts ; three claims did not survive reading both	(i) “train only on public CBFS” is true of the <i>entry</i> and false of the <i>paper</i> , which trains on mixed NASA hump + CBFS — and the hump is a scored case, so the paper’s protocol scored on this benchmark would be in-sample by construction; (ii) “zero-shot generalise to duct” is stated by neither artifact and is <i>our</i> inference from the entry’s training plus the leaderboard, sound but ours to carry; (iii) the rank-2 model is SST-QCRC , carrying an untrained QCR2000 quadratic constitutive term ($c_r = 0.3$) that no laboratory document had noticed.

the “5 of 8” headline; and a citation audit that found two of our own quotations had gone stale *because we improved the pages they cited* (§6).

4.3 The ducts: what is structurally missing

The deficit is the ducts. Rank 2 reaches 0.0455 and 0.0399 on AR_1 and AR_3 where we reach 0.0811 and 0.0775; those two cases were 62.9% of the round-2 gap and remain the dominant term. Four measured findings explain why, and none of them is fixable by more model capacity on the present inputs.

1. **A proven algebraic degeneracy.** Uncorrected RANS produces *exactly* zero secondary flow on every duct: the streamwise fraction of mean $|\mathbf{U}|$ is 1.0000 on all four training cases, and the maximum transverse $|\mathbf{U}|$ is $\sim 1\text{e-}15$ against $O(10)$ streamwise, because a linear Boussinesq closure has no mechanism for Prandtl’s secondary flow of the second kind. For a mean field of that form, I3_S3 and I4_W2S vanish identically — verified on the run’s own data at machine epsilon ($\sim 1\text{e-}14$) and independently over 50,000 random shear-rate pairs. Two of the seven features carry *zero* information on the entire duct family, by mathematical necessity.
2. **And it is worse than we published.** Two further identities were measured and had never been recorded: $\text{I2_W2} = -\text{I1_S2}$ *bit-exactly* (residual 0.0 on all eight duct cases) and $\text{I5_W2S2} = -\text{I1_S2}^2/2$ to 3.5×10^{-16} . **Five of the seven features carry one independent degree of freedom on a duct, not five**; the effective input dimension is 3, or 4 once Variant D’s $d/d_{\max}$ is added, where the physics is three-dimensional. The degeneracy is

a property of the *baseline we are correcting*, not of duct geometry — the DNS field does carry secondary flow. Control: on the four periodic-hill test cases the same residuals run from 2.5×10^{-3} to 9.1×10^{-1} , so the test is not vacuous.

3. **A hypothesis of our own, tested and killed.** Every feature is an $O(3)$ invariant of $\nabla \mathbf{U}$ or a wall distance, all invariant under the $y \leftrightarrow z$ swap that maps the square duct’s quadrant onto itself — which looked like a hard error floor built in by construction. Measured on AR_1_Ret_180, the unreachable component is **0.012% of the correction’s energy**, because the true field respects the same symmetry the features do. Refuted, and shipped refuted.
4. **What binds instead is transfer.** A k -NN conditional-variance bound on the four training ducts puts the reachable error at 0.011–0.027 against floors of 0.069–0.107: in-family, the feature set is not the constraint. The binding constraint is the Re_y extrapolation of §3.1, and **no legal test of Reynolds transfer at a doubled velocity scale exists inside this benchmark** — which is why Variant D is validated for no-harm on every held-out set the rules permit and is *not* proven to close the deficit.

Separately, and stated as a *bound* rather than a decomposition: the missing secondary flow is 2.07–2.22% of bulk velocity in the DNS and identically absent from RANS, so it can explain at most its own magnitude of our duct error — at most 24% on AR_1 and AR_3. **Scaled MAE does not decompose exactly, so this is a bound and not an error budget.** Its consequence is nonetheless directional: at least 76% of our duct error is streamwise-profile error, and rank 2’s *total* duct error (4.55%, 3.99%) is already smaller than our streamwise-only remainder ($\sim 7.0\%$, $\sim 6.6\%$). A scalar eddy-viscosity-style correction — exactly what the field-inversion line targets — is well aimed at the dominant term, not wasted against an anisotropy-only gap.

4.4 What the two entries above us do that we do not

Versus Reissmann, Fang & Sandberg (0.0595, gap 0.0059). Gene-expression-programming symbolic regression of an anisotropic Reynolds-stress correction, fitted concurrently with a differential-evolution refit of the k - ω SST coefficients (lineage: Weatheritt & Sandberg, *J. Comput. Phys.* 325 (2016) 22–37; implementation Reissmann, Fang, Ooi & Sandberg, *GPEM* 26(1):12 (2025), DOI 10.1007/s10710-025-09510-z). An algebraic stress correction inside the PDE *can generate* Prandtl’s secondary flow of the second kind and carries it across Reynolds number through the re-solve — exactly the two axes where they beat us worst. They train on the same duct family we do, AR 1–10 at $Re_\tau = 180$: **the difference is not data, it is model form.** Where we beat them: both alpha_15 cases (0.0501/0.1011 against 0.0592/0.1339) and both declined alpha_05 cases.

Versus Wu & Zhang (0.0624, gap 0.0030). One training flow, a five-constant symbolic law, and a re-solve beat our 21-case gradient-boosted model everywhere outside the periodic-hill family. Their SST-QCRC combines a fixed, untrained QCR-style quadratic Reynolds-stress term ($c_r = 0.3$, “directly adopted”, no data-driven training) with a learned multiplicative field β_{CND} on the ω -destruction term obtained by conditioned field inversion and PySR symbolic regression; the final law is three local features collapsed to $\beta_{\text{CND}} = \max(-0.1157, 0.0058525 \lambda_2) \cdot \lambda_2$, bounded below 4 by hand for stability (Wu & Zhang, *The Training of the SST-QCRC Model*, the entry’s own description document; backing paper Wu, Zhang & Zhang, *AIAA J.* 63(2):687–706 (2025), DOI 10.2514/1.J064416, arXiv:2402.16355). What they do that we do not is **correct causes rather than symptoms** — the ω equation rather than the velocity field — and buy generalisation from compactness. The QCR term supplies duct secondary flow they never trained for.

An external finding worth recording. Per their own paper’s main-text Table 1, the NASA wall-mounted hump — the same flow as the scored NASA_2DWMH — is one of the nine training

flows of the rank-3 entry’s method (Liu, Wang, Zhao & Xiao, arXiv:2509.17189). **Stated as an observation from their own table, not as an accusation:** the paper is not the submission, and whether the leaderboard entry retrained excluding the hump is not determinable from public sources.

5 The field-inversion line: blocked, unblocked, verified

The decomposition points at a scalar eddy-viscosity-style correction inside the equations. That is the FIML family, and reproducing it is the laboratory’s Ladder-B line. Over the last week it moved from a hard blocker to the first converged, verified closure-relevant gradient.

5.1 Capability established, and a deviation disclosed

Stage 1 (dafoam/ladder-b/S1_FIML_FIELD_INVERSION.md, commit 1cd44c04) established the capability on DAfoam v5.0.0 / OpenFOAM v2506 (He, Mader, Martins & Maki, *AIAA J.* (2020), J058853). Four results, in the order the evidence forces:

1. **The mesh-warp defect that fails this laboratory’s shape-optimisation cases cannot reach a β gradient** — measured, not reasoned. A counter monkeypatched onto DAfoamWarper.__init__ and compute_jacvec_product printed {"warper_init": 0, "warper_jacvec": 0} on every field-inversion adjoint that ran to completion. A class never instantiated cannot have a method called on it: the warp is not in the model. FIML is reachable on a code path where shape optimisation here is not.
2. **No custom turbulence library is needed**, and a previous record’s central disclosure was retracted on that basis: betaFI0mega_ and betaFIK_ are members of DAfoam’s own DAK0megaSST, constructed READ_IF_PRESENT with default 1.0.
3. **The disclosed deviation.** DAfoam’s β multiplies the ω equation’s **production** term; the destruction term’s β is the F1-blended model constant and carries no field hook. Anything built on this hook is a production-term inversion and must be recorded as such. **This is a real reduction in what Stage 1 can claim, and it is the honest one: building the production version and calling it the roadmap item would have been the quieter and worse choice.**
4. **The adjoint works and is finite-difference verified** on a 5,000-cell case with 5,000 design variables: 2.674% against the real objective gradient direction at step 10^{-3} and 2.683% at 10^{-4} , with 2.2–3.1% per resolvable component and zero sign flips among components the finite difference can resolve. The residual $\sim 2.7\%$ is *step-independent* (a factor of ten in step moves it by 0.009 percentage points, where genuine central-difference truncation would fall by $\sim 100\times$) and *convergence-independent* (tightening the primal 1000-fold, to an achieved residual of 9.947e-12, moved it from 2.6739% to 2.6728%). It is real, unexplained, and recorded rather than buried.

A finite-difference `check_totals` over a per-cell β field would need $2N$ primal solves — 10,000 on the tutorial case and 103,252 on the hump. Field-inversion gradients cannot be verified the way shape gradients are; the directional-plus-sampled-component protocol is the affordable substitute, and that is what “FD-verified” has to mean at this design-variable count.

A protocol defect found by auditing our own logs. A monitor replay flagged four of 34 finite-difference runs printing DAfoam’s velocity-bound message. The escalation asked which four points to exclude. **Both the question and its premise were wrong.** DAfoam gates that message on `printInterval`, which the campaign ran at the default of 100 — so each log reported

clipping from 1% of its iterations. Re-running twelve points at `printInterval 1` found **689 clip events against the 3 the archive recorded: a $230\times$ undercount**, with eleven of twelve runs clipping including the unperturbed baseline. There was no clip-free subset to retreat to. What the re-run also found is that all twelve objectives return *bit-identical to sixteen significant figures* with iteration counts matching exactly, and all 689 events fall in iterations 27–142 of runs 1815–3291 iterations long. The clip is a startup transient the iteration discards; the verification stands on a narrower and better-stated claim than before — not “no clip occurred”, which was never true and was never measured, but “the clip is a transient and the objective is provably unchanged by it”. The real defect found was in the protocol: the analyser read only the objective line and never inspected the log it came from.

5.2 The blocker, and its mechanism

Both closure-relevant target cases failed identically. The DA Foam/PETSc discrete-adjoint GMRES solve returned `PetscConvergedReason: -9 (KSP_DIVERGED_NANORINF)` at iteration 0 on CBFS (21,000 cells) and on the NASA hump (51,626 cells) — two meshes, two objectives, one signature. Memory was excluded by measurement (host `MemAvailable` bottomed at 18.99 GB with the Jacobian colouring completing normally), as were mesh quality, wall treatment, a non-differentiable velocity clip, primal convergence tolerance, and the objective function.

The mechanism was then located rather than guessed (`S1_FIML_FIELD_INVERSION.md` §4 addendum; `PROOF.md` §25.2–25.3):

- **The whole matrix-ordering axis was run.** Of DA Foam’s five orderings, two (`rcm`, `1wd`) produce the NaN and three (`natural`, `nd`, `qmd`) produce honest stagnation at -3 with the residual flat to 13 significant figures; *none* converges. The initial residual is identical (7.091590452305e-04) in all five, as it must be. A single buggy ordering would not do that; a factorisation hitting a destructive zero pivot under some elimination orders and not others does exactly that.
- **The cause was reproduced with no DA Foam in the loop.** The preconditioner matrix and right-hand side were dumped with stock PETSc runtime flags and analysed offline. The matrix (210,592 square, 13,710,468 nonzeros) has no zero rows, columns or diagonal entries. `scipy.sparse.linalg.spilu` returns “*Factor is exactly singular*” — the -9, in a second independent implementation — while `splu`, full LU *with partial pivoting*, solves the same system to $\|Ax - b\|/\|b\| = 2.5 \times 10^{-12}$. The system is nonsingular and consistent. **Fill adds fill, not pivoting**, which is why raising the ILU fill level had never helped.

5.3 The unblock, and what it cost

`dafoam/ladder-b/w4_ADJOINT_PC_UNBLOCK.md` (commit 1cd44c04) closes it. The runtime-options route was tested first and found *reachable but empty*: `sub_`-prefixed factor options do land in the deployed factor (a 1e-8 zero-pivot threshold is visible in `PCView`), correcting a standing belief that nothing reached the sub-preconditioner — and every one of them still returns -9, so the rebuild was justified by measurement rather than convenience.

The fix is one word. DA Foam hard-codes `PCType localPCType = PCILU` for the ASM sub-blocks; a single-hunk patch makes it `PCLU` when `DAFOAM_SUBPC_TYPE=1u` is set in the environment, and changes nothing otherwise. Results:

- **CBFS converges** — `PetscConvergedReason: 2`, 667 iterations, 243 s at 4 ranks — on B3’s exact -9 configuration, changed only by the environment variable. The residual history is instructive: the first 400 iterations crawl, because the assembled matrix the LU inverts is not the matrix-free operator GMRES applies, and then the residual falls six decades in 267 iterations. **Slow-then-superlinear is what a correct strong preconditioner on**

a mismatched operator looks like; flat to thirteen digits is what the dead ILU looked like. They are not the same thing.

- **The first closure-relevant field-inversion gradient exists:** 21,000 per-cell β components on CBFS, objective $\text{varianceU} = 1.5279278906359758 \times 10^{-2}$, gradient norm $1.4558\text{e-}05$.
- **The regression control was run before the treatment:** the patched image with the switch *unset* reproduces the -9 at iteration 0 with all 13 residual digits identical. The rebuild is behaviour-neutral by default.
- **On the hump the same switch moves the failure from catastrophic to honest:** the factorisation completes and the residual descends monotonically ($1.094 \rightarrow 0.954$ over 900 iterations, accelerating) but does not converge, and the run was stopped deliberately when host MemAvailable fell to 1.62 GB on a shared box. **The hump blocker is no longer a singular factorisation — it is Krylov convergence rate against a memory envelope**, which is a different, honestly named problem.

Finite-difference verification of the CBFS gradient met its pre-declared gate at 13–45 times the required tightness (Table 6), with the unperturbed control run reproducing the adjoint run’s objective to *exactly zero difference* across all 17 digits. Total cost, billed at full occupancy for full wall-clock including a 600s ordering timeout: 185.0 core-min against a 240 core-min budget.

Table 6: Finite-difference verification of the CBFS β gradient (21,000 design variables). The last row was produced by an independent verification agent at a cell the laboratory never published.

Cell	h	Central FD	Adjoint $g[i]$	Rel. error	Source
5491 (largest $ g $)	0.05	1.914384790951e-06	1.916018813330e-06	0.085%	W4 §5d
5491	0.10	1.907246786675e-06	1.916018813330e-06	0.460%	W4 §5d
6740	0.05	1.679642395377e-06	1.678653382471e-06	0.059%	W4 §5d
12486	0.05	1.472401682783e-06	1.469472928907e-06	0.199%	W4 §5d
6490	0.05	1.520027166028e-06	1.519705840939e-06	0.021%	supervisor sweep

The doubling of the error from $h = 0.05$ to $h = 0.1$ on cell 5491, with the finite difference moving *away* from the adjoint value as the step grows, is central-difference truncation behaving exactly as $O(h^2)$ predicts: the agreement is step-limited, not noise-limited, which is the signature of a correct derivative measured against a resolvable objective.

5.4 Independent adversarial verification

The claim was then attacked by a separate verification agent working under the doctrine that the claim is wrong until it survives attack (`dafoam/VERIFICATION_cbfs_unblock_supervisor_sweep.md`, commit `9c19ccc8`). Nothing was taken on the record’s word:

- The patch was diffed *against the built image*, not against the patch file. An md5 sweep of every `.C/.H` in the DAfoam source tree of both images shows exactly two changed files — `DALinearEqn.C` and its `lnInclude` copy — and all three AD-mode libraries contain the new environment-variable string, so all three were rebuilt from patched source.
- The finite-difference table was recomputed from the raw perturbation vectors: each perturbation file was verified to differ from the all-ones baseline at *exactly one index*, by exactly $\pm h$.
- A new finite-difference point was derived at cell 6490 — rank 10 of $|g|$, a cell the laboratory never chose — in an isolated case copy with the sweep’s own driver: 0.0211%, tighter than any published cell.

- The env-off regression was re-run cold, independently, reproducing -9 with all 13 residual digits.
- The cost ledger was reconciled against log timestamps to the second (693.04s measured against 693s billed; 642s summing exactly).

Verdict: no defect found, five of five attacks confirmed, one disclosed caveat restated rather than allowed to go quiet — the hump attempt of W4 §5b computes its objective against NASA_2DWMH experimental data, a scored case; it produced no gradient, nothing from it feeds any entry, and the gate claim rests on CBFS alone.

5.5 What the rank-2 method actually is, and what a reproduction must copy

The W2 reading (`campaign/W2_WU_ZHANG_DESTRUCTION_FIML_READING.md`, commit 393ad74f) read both Wu artifacts in full and returned with the correction of Table 5 plus an implementation specification. Its operational consequences are three:

1. **A reproduction must copy the *entry's* protocol, not the paper's.** The paper's models train on mixed NASA hump and CBFS inversion data; the hump is a scored case, so the paper's protocol scored on this benchmark is in-sample by construction. The entry's CBFS-only choice is what keeps a reproduction clean.
2. **A β -on-destruction patch alone reproduces the paper's inversion capability, not the deployed rank-2 model.** Duct parity additionally needs the untrained QCR2000 constitutive term, filed separately as `w3-qcr-constitutive-term-for-rank2-parity`. Any score from a β -destruction-only model must not be set beside 0.0455/0.0399 as like-for-like.
3. **The patch was specified from the installed source, not from documentation.** The ω equation is assembled in two places (`DAkOmegaSST.C:745` in `calcResiduals()` and `:870` in `getFvMatrixFields()`) and both carry the identical destruction line; patching only one would leave the preconditioner assembling a different equation than the residual — on a stack whose standing blocker *was* a factorisation of exactly that matrix, an inconsistent preconditioner would be a new confound indistinguishable from the old one. Seven named risks are recorded with the specification, including that the conditioned shielding function uses the constant 8 from the paper where OpenFOAM's built-in DDES shield would silently supply 20.

5.6 What is running now

The Stage 1 CBFS inversion is pre-registered and under way (`dafoam/ladder-b/S1_CBFS_INVERSION_PREREGISTRATION.md`, commit e6321e95), written before any inversion iteration ran and while the only existing numbers were the baseline objective and the single verified gradient at $\beta = 1$. It fixes, in advance: the loss $J(\beta) = \lambda_{\text{QoI}} \text{varianceU}(\beta) + \lambda_{L2} \sum_j (\beta_j - 1)^2$ with $\lambda_{\text{QoI}} = 65.448114804931393$ so that the data term equals exactly 1 at $\beta = 1$ (Wu & Zhang's own normalisation) and $\lambda_{L2} = 10^{-4}$ declared at the top of their stated band and *not* tuned; bounds $[0.2, 4.0]$, the upper bound theirs and the lower bound ours by choice and disclosed as such; L-BFGS-B rather than their SLSQP, because SciPy's SLSQP would allocate a dense 3.749×10^9 -double workspace — 30.0 GB, the whole of this host's RAM — at 21,000 design variables; a hard budget cap of 600 core-min with a stop rule that records a capped run as *capped, not converged*; and two success criteria, a 30% reduction in the data term and a requirement that more than half of the top-decile β deviations fall inside a pre-declared spatial window around the recirculation bubble and separated shear layer.

It also fixes its own label before any result exists: **this is a production-term inversion**, at ~ 36 evaluations against Wu & Zhang’s ~ 140 SLSQP iterations, with a field-variance loss rather than their 30 sparse points — three independent reasons no number it produces may be set beside theirs as like-for-like.

6 Verification and integrity

6.1 Leakage guards, in code rather than in prose

The one strict rule was audited line by line in the code rather than taken from our own documentation (`CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md` §4.1):

- Ground-truth reads occur only inside loops over the 21 training cases. The test-case loop calls the RANS-field loader only, annotated `# no U_LES read`.
- The split is enforced by executable assertions, not comments: `assert len(_PH_TRAIN) == 21 and len(_PH_VAL) == 4`, and assertions that the training, validation and test name sets do not intersect. The test list is imported *solely* to assert non-intersection.
- The CSV exporter does not merely omit scoring calls: before any pipeline work it replaces `score`, `score_from_csv`, `evaluate_by_case`, `evaluate_from_csv_by_case`, `evaluate_individual_case` and the ground-truth accessors with raising stubs across three namespaces, then *proves the guard is armed* by calling `score()` and catching the refusal before continuing. If a future edit reintroduces a scoring call, the script dies rather than quietly spending one.
- An in-sample gate (`closure_in_sample_gate.py`) runs before and after closure work and passes.

The residual soft leakage, disclosed by us before anyone asks. Across rounds, test feedback influenced *which machinery got built*. The round-3 record states it in its own words: “*The gate’s MOTIVATION came from round 2’s official call . . . That is soft, adaptive leakage, and it is real.*” The defence is that the gate’s every parameter derives from 21 training cases and was reproduced byte-for-byte from train-only data before touching test features; that seeing round-2’s scores changed *when* it was applied, not *what* it outputs, so its four test decisions would have been identical had round 2 never been scored; and that the benchmark itself ships the ground truth and instructs submitters to preview per-case scores, so an interpretation under which observing them is a violation would make the benchmark’s own documented workflow a violation. **Verdict: not a violation of the rule as written; residual risk low but non-zero, and interpretive rather than factual.** It belongs in the submission’s cover material so that a reviewer *finds* it written down rather than discovers it.

6.2 The scoring-call ledger

Five distinct prediction sets have ever been scored against the eight test cases’ ground truth: the RANS-identity floor, and rounds 1 through 4. The unit counted is *distinct prediction sets*, which is a defensible and meaningful unit — and the compliance audit caught a docstring claiming something stronger and demonstrably false (“exactly one call” where the same run makes four invocations across two prediction sets, the extra pair re-scoring the already-counted floor as a harness check). The claim was substantively harmless and was fixed anyway, because **an entry whose entire credibility rests on precise self-accounting cannot ship with a sentence a reviewer can falsify by reading forty lines further down the same file.**

The ledger is a self-imposed discipline. The benchmark imposes no limit and instructs submitters to preview their score. No public statement of ours may imply otherwise.

6.3 The adversarial methods audit

An audit conducted on 2026-08-04 proof-checked the closure line end to end under the instruction to assume the records overstate until verified, test-blind throughout (`CLOSURE_METHODS_COMPARISON.md`, commit `ac2f37ee`). The metric definition held as written. Six gaps were found in our own evidence, ranked by how much the entry’s credibility rests on the unverified assumption, each costed as a proposal:

- G1 — seed sensitivity, real and live.** `random_state=0` everywhere and no model ever re-trained under another seed. This is not pedantry: at 327,600 training cells the PH model crosses scikit-learn 1.9.0’s 200,000-row binning subsample, so the bin edges — hence the trees, hence three of the eight predictions — depend on the seed, against a gap to rank 2 of 0.0030.
- G2 — no physicality check on any corrected field.** $\nabla \cdot \mathbf{U}$ never measured, on the one dimension every other entrant gets by construction.
- G3 — baseline convergence evidence exists but is uncited** for all eight predictions. It matters twice over on the two declined cases, where the shipped solve *is* our submission. **Closed 2026-08-05** (`CLOSURE_CHALLENGE_STATUS.md` §4b, commit `11ff63b1`): the four periodic-hill test cases carry final residuals of $\sim 1\text{e-}10$ after 20,000 iterations, and a sweep of all 29 cases reproduces the pattern on 28 of 29 — with the exception recorded honestly rather than smoothed, one *training* case whose shipped residual file is truncated at iteration 76, so that its convergence is plausible but not evidenced. The three duct test cases terminate with “*SIMPLE solution converged*” at 405 / 1540 / 7009 iterations under explicit `residualControl`, and the $O(0.4)$ – $O(0.7)$ transverse residuals that look alarming are recorded as a normalisation artifact on a machine-zero field ($\max |U_y| = 2.1 \times 10^{-15}$ against U_x up to 108 m/s), noted so that nobody later mistakes them for a divergent solve.
- G4 — manifest gap.** The three new round-4 duct CSVs carried no recorded SHA-256 and the round-4 submission directory had no manifest; the hashing discipline covered only the five copied files. **Closed 2026-08-05** (`closure_challenge_submission_round4/MANIFEST.json`, commit `30fd6d7a`): all eight files hashed, zero scoring calls, guaranteed by a script that never imports the benchmark package at all.
- G5 — gate statistical power** is thin and already at its disclosed limit; widen from 4 held-out decisions to 25 by leave-one-out over all periodic-hill cases.
- G6 — no training-convergence curve** for either gradient-boosted model.

None requires a scoring call; none touches test ground truth; all six keep the leakage guards armed. Four are now closed: G1 and G2 were taken first, on pre-registered thresholds (§6.4), and G3 and G4 landed on 2026-08-05.

6.4 Seed stability and field physicality: the lines drawn before the measurements

The pre-registration for G1 and G2 was committed (`closure_challenge_stability_physicality_audit.md` §0, commit `73fa6a33`; results at `fee5eaf9` and `cbe040f6`) *before* either run existed, so the materiality lines cannot have been drawn around the results:

- **G1:** retrain the PH model at eight seeds with everything else frozen, score on the four validation cases only, and compute a test-side prediction-spread bound requiring no ground truth at all. *Material if* the overall-equivalent induced spread reaches 0.0003 (10% of the gap to rank 2) by either estimate, *or* if any single case’s spread reaches 0.0030 — a one-case spread the size of the whole gap being material regardless of the mean.

- **G2:** push both the raw RANS and the corrected field through the *identical* already-validated Green–Gauss operator so that discretisation error cancels in the comparison, and report the volume-weighted RMS divergence and its ratio. *A ratio of 2 or more on any model-corrected case is recorded as a material physicality cost of the method class and must be carried into the submission’s disclosure section; a ratio below 2 is still reported per case, quantified, with no editorialising either way.*

G1, measured. The results are recorded in §1 of the same audit file (commit `fee5eaf9`), with raw numbers in `closure_challenge_seed_sensitivity.json` (generated 2026-08-04T18:55Z; zero scoring calls; guard re-armed and proven armed first; ground truth read for the 21 training and 4 validation cases only). Seed 0 reproduces the recorded round-1 model exactly (validation pooled 0.0876) and its predictions match the shipped round-4 CSVs to 5e-10. Across seeds 0–7 the pooled validation metric spans only 0.000276, and the two proxy ranges for the served `alpha_15` predictions are 0.000568 and 0.000520; the overall-equivalent proxy estimate is $S_{\text{proxy}} = 0.000261$, just below the 0.0003 line. But the test-side spread *bounds* — the largest disagreement between any two seeds’ own predictions at the 1000 evaluation points, normalised the same way the metric is — reach **0.0085** and **0.0099** on the two `alpha_15` cases and 0.0010 on the hump, giving $S_{\text{bound}} = 0.00242$.

Verdict: material, by two of the three pre-declared triggers — $S_{\text{bound}} \geq 0.0003$, and two single-case bounds at or above 0.0030 — and both halves of the reading belong in the record:

- **Where truth is legitimately available, the metric is stable under seed.** On the four held-out validation cases the per-case spread is 0.0005–0.0012 and the overall-equivalent is ~ 0.00026 , a tenth of the gap to rank 2. That is the best available *estimate* of the seed effect on the score, and it is small: the entry gains a measured rather than assumed stability property, an estimated one-seed uncertainty on the overall of about 0.0003.
- **Where truth is not available, the only honest statement is the bound, and it is not small.** Across seeds the trees move the pointwise test predictions by 0.85–0.99% of the velocity scale on the two served `alpha_15` cases — per-case movement up to roughly three times the entire gap *if it failed to cancel against the truth*. On validation cases that wobble demonstrably cancels to the 0.0006 level in the metric; expecting the same on test is reasonable, but it is an extrapolation and not a measurement, and the pre-registered rule counts the bound. Tightening it would cost a scoring call and is not taken.

The consequence was fixed in advance and has been executed: **our leaderboard-gap language now carries a seed-uncertainty qualifier** in the status record and on the public page. Scope is exact — the duct model behind three predictions is bin-deterministic at 41,971 cells, so `random_state` is inert for it by construction rather than by measurement, and the two declined cases are the organisers’ own solve, so the five non-PH predictions contribute zero seed variance and the division by eight is exact.

G2, measured — and the finding is against us. The divergence audit completed on 2026-08-05 (310 s on the two-core cap, zero scoring calls, 37 cases; results in §2 of the audit file, raw record `closure_challenge_divergence_audit.json`). Every regenerated test field was matched against its shipped round-4 CSV *before* its divergence number was believed — maximum absolute difference 4.8e-10 to 4.9e-8, all at CSV write precision — so the numbers are about the submitted fields and not a lookalike.

Verdict: material. Thirty-five of thirty-six model-corrected cases sit at or above the pre-registered ratio of 2, and five of the six corrected submissions exceed it; only `NASA_2DWMH` (1.53) stays under, for the consistent reason that the PH-trained correction is small relative to that case’s own gradient scale — the same model that barely moved its score barely moves its continuity

Table 7: Volume-weighted RMS divergence of the eight submitted fields, both fields through the identical validated Green–Gauss operator. The scale-free column is the continuity error relative to the field’s own velocity-gradient magnitude, $\|\nabla \cdot \mathbf{U}\|/\|\nabla \mathbf{U}\|_F$, given for the RANS field and for the submitted field; the RANS value is the operator-consistency floor, since the solver enforces continuity on face fluxes rather than on reconstructed cell gradients.

Case	Submitted field	Ratio	rel. RANS	rel. corrected
alpha_15_13929_4048	PH-corrected	58.0	0.18%	10.5%
alpha_15_13929_2024	PH-corrected	123.6	0.08%	9.7%
alpha_05_4071_4048	declined (raw RANS)	1.000	0.47%	0.47%
alpha_05_4071_2024	declined (raw RANS)	1.000	0.27%	0.27%
AR_1_Ret_360	duct-D corrected	$\sim 1\text{e}15^\dagger$	2.4e-17	2.3%
AR_3_Ret_360	duct-D corrected	$\sim 4\text{e}14^\dagger$	6.9e-17	3.1%
AR_14_Ret_180	duct-D corrected	$\sim 7\text{e}14^\dagger$	4.6e-17	3.4%
NASA_2DWMH	PH-corrected	1.526	0.43%	0.66%

[†] The duct ratios are divisions by machine zero and should be read as such: a fully developed, exactly unidirectional RANS duct field is cell-wise divergence-free to roundoff (9e-12 against a gradient scale of 4×10^5). The interpretable statement is the last column, not the ratio.

error. The train and validation cases show the same corrected-field levels (8–19% on hills, 2–3% on ducts), so this is a property of the method on its own turf and not a test-set artifact.

Three things follow, and none of them is an excuse:

- **The cost is real and now has a number.** The correction takes fields whose measured continuity error sits at the operator floor and returns fields violating continuity at $\sim 10\%$ of the gradient scale on the two corrected hills and 2.3–3.4% on the three ducts. That is far above the operator floor, so it is the field and not the operator.
- **The scoring metric never sees it.** Nothing here changes 0.0654. But it is exactly what a referee of the description document will ask, and per the pre-registered consequence it is carried into the submission’s disclosure section — “we measured it and here is the number” beats silence, even when the number is against us.
- **The two declined cases are, again, the strongest rows.** Their ratio is exactly 1.000 because the gate’s “off” state ships the organisers’ own solve and inherits its physicality untouched. The one part of our submission that is physical by construction is the part where our model does not appear.

This is also the measured case for the field-inversion line of §5. The argument for moving the correction *inside* the equations had been an argument from taxonomy and from the duct decomposition; it is now an argument from a number on our own submitted fields. A correction that re-enters the PDE and is re-solved returns a field satisfying continuity because the solver enforces it — which is precisely what the four entries above and below us get for free, and precisely what Stage 1 is being built to give us.

6.5 The pre-registration that said no

The clearest evidence that the discipline binds is the round-5 candidate (`closure_challenge_R5_ALPHA05_REGIME_PREREGISTRATION.md`, commit `be642415`). C2 had recommended training a second, regime-specific model as strictly better than gating, because it recovers the term instead of merely declining to lose it. The candidate was built — 11 of 21 training cases qualified under a membership rule defined by the same frozen C1 gate that makes the test-time decision, and the

gate refit reproduced $\alpha = 0.7499$ and threshold 0.1263 before membership was computed, the script exiting rather than defining membership from a drifted gate.

The go/no-go rules were declared in the committed training script’s docstring before any leave-one-out delta existed. Measured:

Table 8: The R5 pre-declared gate, and what the measurements did to it.

Rule	Measured	Verdict
(i) the validation case improves	alpha_05_10071_4048 : 0.0759 $\rightarrow$ 0.0447 (−0.0312), where the global round-1 model gives 0.0772, a hurt	PASS
(ii) mean leave-one-out delta < 0	−0.0195 over 10 cases, 8 of 10 improved, two by more than 0.035	PASS
(iii) no case hurt by more than +0.010	alpha_05_4071_3036 : 0.0492 $\rightarrow$ 0.0659, +0.0167	FAIL

Verdict: NO-GO. The rules were not revisited after the numbers, the scoring call was not authorised, and nothing was scored. Round 4 stands as the entry of record.

The finding underneath is worth more than the candidate would have been (recorded as lesson L-33). The case that fails is **alpha_05_4071_3036** — the training sibling of both declined test cases, sharing their α and their 4071 parameter and differing only in the third index — and it is the lowest-baseline case in the entire training split (0.0492). The leave-one-out table splits cleanly on baseline quality: every case with baseline ≥ 0.058 improves and both cases below ~ 0.055 get worse, in both variants. **C2’s mechanism is recursive: train only on the low-baseline regime and the hurt-where-RANS-is-already-good boundary does not disappear — it moves down, from ~ 0.07 – 0.13 to ~ 0.055 , and reappears inside the regime.** A correction model fitted to a population still damages the best-baseline members of that population.

A successor candidate — a second-stage threshold declining even the regime model below ~ 0.055 — is obvious and is explicitly *not* adopted, because it was conceived after those numbers existed and adopting it now would be exactly the rules-revisited-after-the-fact move the document’s own §3 forbids. It is filed as a recommendation needing its own pre-registration, its own membership audit, and a held-out design that does not lean on a single in-regime validation case.

6.6 Corrections and withdrawals register

- **The published duct diagnosis was wrong** and was corrected by the round that acted on it (§3.1).
- **“No alternative training-family model exists for NASA_2DWMH” was withdrawn:** CBFS — the curved backward-facing step, the nearest flow in this benchmark to a wall-mounted hump — ships as *training* data with its LES fields and is legal to fit on. The pre-registered argument that chose the periodic-hill model was a two-way comparison of hills against ducts in which CBFS is not named, not rejected and not mentioned: the choice was defended against the worse of two candidates while the better third sat unused in the training set. What still keeps the case shut is not the absence of a candidate but *who* may choose and *when* — anyone choosing today has seen round 2’s score, so taking that route legally requires a pre-registration frozen first. Both routes are priced; neither has been taken.
- **“We are last on the ducts” was never true.** A working belief repeated in internal notes was re-derived from the harness and found false; after round 4 we are last on NASA_2DWMH alone.

- **Two of our own quotations went stale because we improved the pages they cited.** A citation audit extracted all 28 quotations in the submission draft mechanically and matched them against normalised source text. Both defects were in citations of sources *we control* — not one was in a citation of the benchmark’s rules, which had been transcribed carefully because they were known to matter. The audit closes the question that no sentence in that document is attributed to a source nobody here has read, and explicitly does *not* close the general problem: nothing checks staleness automatically, so it must be re-run at the moment of approval, not before.
- **An over-generalisation in our own solver record was corrected on measurement:** a prior finding that a certain PETSc failure code fools DA Foam’s success gate was found to be specific to -5; -9 never fools it. It matters in the safe direction — the blocked cases were never at risk of silently returning a wrong gradient.

7 Outlook

7.1 The technical path

Duct parity is the largest available term. Matching the best published entry on the two aspect ratios we trail is worth about 0.011 on the overall — more than the entire 0.0059 gap to rank 1. Every move currently visible in that family is either contaminated by knowledge of per-case test outcomes or needs evidence this benchmark cannot supply, so a structural route has been filed instead: the scalar-invariant feature set is provably three-dimensional on a duct while the tensor basis is not. Measured on two duct training cases, the second basis tensor carries $1.87\text{--}2.36\times$ the norm of the first, with 71% of its magnitude and 91% of the third tensor’s in the transverse block that generates secondary flow. That is a specific structural argument for the duct family belonging to a tensor-basis ansatz.

Rank-2 parity needs two components, not one. The β -on-destruction model patch is specified line by line against installed source and filed; the untrained QCR constitutive term is filed separately. Landing only the first reproduces the paper’s inversion capability, not the deployed rank-2 model, and no score from it may be compared like-for-like.

Stage 1 is running. The CBFS production-term inversion is pre-registered, budgeted at 600 core-min, checkpointed every evaluation, and labelled in advance. It can establish that the verified gradient drives an actual descent on a closure-relevant case, where a production-term β wants to move on CBFS, and the real per-iteration price that Stage 2 planning needs. It cannot establish anything about destruction-term physics, about generalisation, or about any benchmark score — and it says so before it runs.

The hump remains open, honestly named. Its blocker is now Krylov convergence rate against a memory envelope rather than a singular factorisation. A Richardson-wrapped configuration capping the restart basis is staged and unrun, awaiting a fresh budget.

Audit items. Four of the six are closed: G1 and G2 both returned material findings against pre-registered thresholds, and G3 and G4 landed on 2026-08-05. What remains is G5, which widens the gate’s evidence base from 4 held-out decisions to 25 by leave-one-out over all periodic-hill cases, and G6, which is free if folded into the retraining runs.

7.2 Submission status

The package exists and is audited. The eight CSVs are written, 1000 rows by 3 columns, comma-delimited and header-free, verified finite and round-trip accurate below $5e-8$, with mean velocity magnitudes falling inside the band spanned by all four accepted submissions on all eight cases

— a check that would catch a point-ordering, column-ordering or units blunder at zero scoring cost. Both defects the compliance audit found (the false docstring; the non-existent submittable artifact) are closed, the evaluation package is cited by commit hash rather than by its misleading version string, and the flagged preprint quotations have been verified verbatim.

The description document’s disclosures are fixed and non-negotiable: the method class and its weakness relative to the family it is ranked against; that two submitted fields are the uncorrected baseline and any leaderboard-best status there belongs to the baseline; the full adaptive-leakage history in our own words; the executable train/validation/test assertions; that the hump regression was not reverted and why; the duct feature degeneracy as a negative result we would rather publish than have inferred from our scores; the scoring-call count correctly framed as self-imposed; the measured continuity cost of §6.4 and the measured seed-uncertainty qualifier on the rank-2 gap; and the prior art on classifier-controlled corrections, with no sentence presenting confidence-gated correction as novel.

What remains is not technical. Two items are outstanding and both are Katie’s: the author line and reference URL, and the proofread and approval without which nothing moves. Two questions are drafted for the steward and unsent — whether a company name is acceptable on a leaderboard row, and whether any terms of use apply to a commercial entity training on data whose repositories carry no licence file at all. **As of today the only thing between this package and the steward is Katie.**

A Provenance index

Table 9: Repository artifacts cited in this report. Paths are relative to `demo-output/website/` unless otherwise shown; commits are those at which the cited text stands.

Artifact	Commit	What it carries
<code>CLOSURE_CHALLENGE_STATUS.md</code>	<code>b9cb56a2</code>	Rounds 1–4 history; the metric; per-case tables; feature-expressivity and generalisation studies
<code>CLOSURE_METHODS_COMPARISON.md</code>	<code>ac2f37ee</code>	Metric read from the evaluation code; entrant dossiers; the dimension-by-dimension comparison; gap findings G1–G6
<code>closure_challenge_C2_error_decomposition.md</code>	<code>1da013a8</code>	The error decomposition and its corrections
<code>CLOSURE_CHALLENGE_SUBMISSION_DRAFT.md</code>	<code>15530f97</code>	Policy, eligibility, compliance audit, draft package, addenda
<code>CLOSURE_CHALLENGE_PRIOR_ART.md</code>	<code>15530f97</code>	Prior art on classifier-controlled corrections
<code>closure_challenge_C1_error_estimator.md</code>	<code>5719374e</code>	The decline gate: fit, validation, and its stated statistical limits
<code>closure_challenge_C6_hump_decision.md</code>	<code>1f5e2403</code>	Both hump routes priced; neither taken
<code>closure_challenge_R5_ALPHA05_REGIME_PREREGISTRATION.md</code>	<code>be642415</code>	The round-5 candidate and its NO-GO

Artifact	Commit	What it carries
<code>closure_challenge_stability_physicality_audit.md</code>	73fa6a33 (§0), cbe040f6 (results)	G1/G2 thresholds committed before either run; results §1–§2
<code>closure_challenge_seed_sensitivity.json</code>	cbe040f6	G1 raw record: 8 seeds, spreads, bounds, verdict
<code>closure_challenge_divergence_audit.json</code>	cbe040f6	G2 raw record: 37 cases, RMS divergence before and after correction
<code>closure_challenge_submission_round4/MANIFEST.json</code>	30fd6d7a	All eight submission CSVs hashed; closes G4
<code>closure_challenge_trained_entry_round4_duct.json</code>	4c2f3562	The entry of record
<code>closure_challenge_duct_reynolds_transfer.json</code>	d0992da2	The round-4 pre-registration
<code>dafoam/ladder-b/S1_FIML_FIELD_INVERSION.md</code>	1cd44c04	FIML capability, FD verification, the clip audit, the blocker
<code>dafoam/ladder-b/W4_ADJOINT_PC_UNBLOCK.md</code>	1cd44c04	The sub-block LU unblock and the first closure-relevant gradient
<code>dafoam/VERIFICATION_cbfs_unblock_supervisor_sweep.md</code>	9c19ccc8	Independent adversarial verification of that claim
<code>dafoam/ladder-b/S1_CBFS_INVERSION_PREREGISTRATION.md</code>	e6321e95	The inversion now running, pre-registered
<code>campaign/W2_WU_ZHANG_DESTRUCTION_FIML_READING.md</code>	393ad74f	Both Wu artifacts read in full; the patch specification

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